

When Healthy-Only Alarm Calibration Does Not Transport Across Permanent-Magnet Synchronous Machines

Online Resource 1 for journal-neutral review - anonymous review copy

The development method was chosen with development fault labels. The PMSG protocol, selected method, feature schema, healthy roles, windows, alpha, and failure gates were then frozen before a PMSG signal variable was deserialized. All later session-anchor analyses are post-reveal. A repository size query had already fetched some Git objects containing MAT blobs, but no signal array was checked out for analysis, loaded, plotted, or summarized; the precise claim is therefore a signal-unrevealed metadata freeze.

Artifact	SHA-256
development features	c3e7e13919a8868580bd3af8c501d8f022600f5a2637460470479ad7faff4743
development protocol.json	a37d38937aee35ae0ed44c5c139a4e0daf54cd85a03f1e105b8cc6ab5f071ec9
development selection	1cccd12c8d7b8d4c6ad40b0a96ebd46ec839739b84b30512a8d4b7ca77324bf5
confirmation protocol	4fc0d12e1e69c09776f8c0f691eddf38552bdf2f7512dc71c426103cc349bfd9
confirmation inventory	4890f864e40fc3345f5af37364a6acf7d854af4ab94b57ebc73e2def94a36914
confirmation features	78bbc520ada5bcfe6f3dc419968e64892aaba48743f5976443cdf4f136346040
confirmation primary	f119fde0c7bc6625abc43d68769602305bf24e0e949a3f7c5e301b36634a63cb

The PMSG source is tag v1.1.0, commit e02fba475cf82b375412a7382143dc29da5241ef, tree 65430929e4886935d463f1e1410b5476d255f4d4. All 225 MAT files passed the common-schema and time-base audit without a compatibility patch.

The detector uses time and three phase currents. Development additionally loads commanded speed and filename load as exogenous context; confirmation uses filename speed and torque setting code. Fault current, relay state, measured angle/speed, voltage, dq quantities, and absolute current scale are excluded. The 26 ordered outcomes are:

Index	Feature
1	sequence_unbalance
2	fundamental_amplitude_cv
3	phase_rms_cv
4	clarke_radius_cv
5	zero_sequence_ratio
6	spectral_entropy
7	sideband_lower_ratio
8	sideband_upper_ratio
9	harmonic_2_ratio_mean
10	harmonic_2_ratio_max
11	harmonic_3_ratio_mean
12	harmonic_3_ratio_max
13	harmonic_4_ratio_mean
14	harmonic_4_ratio_max
15	harmonic_5_ratio_mean
16	harmonic_5_ratio_max
17	thd_2_to_5_mean
18	rms_ratio_a
19	crest_a
20	kurtosis_a
21	rms_ratio_b
22	crest_b
23	kurtosis_b
24	rms_ratio_c
25	crest_c

Index	Feature
26	kurtosis_c

PMSG feature inventory: 117 standalone-health windows, 648 pre-fault windows, 432 active-fault windows, and 1,296 recovery windows, totaling 2,493 rows from 225 records. Only t , I_a , I_b , and I_c were deserialized.

Values below use only the six interpolation loads (5-30 N m) for selection. Each method has 48 healthy test blocks and 288 fault blocks. Detection is the equal-record macro average of actionable block alarms. Development fault labels select the method; these rows are not external confirmation.

Method	Healthy alarms	FAR	Detection	Abstention	Eligible
Unconditioned residual	1/48	2.08%	28.47%	0.00%	yes
Linear residual	1/48	2.08%	76.39%	0.00%	yes
Quadratic residual	1/48	2.08%	81.25%	0.00%	yes
Spline residual	1/48	2.08%	82.29%	0.00%	yes (selected)
Spline + local scale	10/48	20.83%	93.75%	0.00%	no
Isolation Forest	1/48	2.08%	65.97%	0.00%	yes
MinCovDet	1/48	2.08%	7.64%	0.00%	yes

All methods use the same 39-window healthy calibration role and 216 fault sessions. A pre-fault session alarms if any of its three frozen pre-fault windows alarms; a fault record is detected if either of two active windows alarms. Only the spline residual is confirmatory. No method passed the 5% FAR point gate.

Method	Threshold	Pre-fault alarms	FAR (95% Wilson)	Detected	Detection (95% Wilson)	AUROC
Unconditioned residual	32.1594	111/216	51.39% (44.76%-57.97%)	173/216	80.09% (74.26%-84.87%)	0.7602
Linear residual	85.0129	57/216	26.39% (20.96%-32.64%)	148/216	68.52% (62.05%-74.34%)	0.7661
Quadratic residual	36.5397	86/216	39.81% (33.52%-46.47%)	164/216	75.93% (69.80%-81.14%)	0.7787
Spline residual [selected]	55.3097	71/216	32.87% (26.95%-39.39%)	156/216	72.22% (65.90%-77.77%)	0.7799
Spline + local scale	120.6509	56/216	25.93% (20.54%-32.15%)	139/216	64.35% (57.76%-70.44%)	0.7486
Isolation Forest	0.5408	90/216	41.67% (35.29%-48.33%)	149/216	68.98% (62.53%-74.77%)	0.7294
MinCovDet	2514.0028	27/216	12.50% (8.73%-17.58%)	122/216	56.48% (49.81%-62.92%)	0.7744

Each speed-by-setting cell contains 24 sessions. These repetitions come from one physical PMSG and are not population replicates. Setting codes 52, 64, and 80 are not asserted to be N m.

Facet	Level	Records	Detection	Pre-fault FAR
fault_family	turns	108	53.70%	30.56%
fault_family	windings	108	90.74%	35.19%
speed_rpm	1200	72	59.72%	29.17%
speed_rpm	1500	72	84.72%	50.00%
speed_rpm	1800	72	72.22%	19.44%
torque_setting_code	52	72	83.33%	48.61%
torque_setting_code	64	72	70.83%	30.56%
torque_setting_code	80	72	62.50%	19.44%
speed_by_torque	1200 / 52	24	75.00%	50.00%
speed_by_torque	1200 / 64	24	62.50%	20.83%
speed_by_torque	1200 / 80	24	41.67%	16.67%
speed_by_torque	1500 / 52	24	91.67%	66.67%
speed_by_torque	1500 / 64	24	83.33%	41.67%
speed_by_torque	1500 / 80	24	79.17%	41.67%
speed_by_torque	1800 / 52	24	83.33%	29.17%
speed_by_torque	1800 / 64	24	66.67%	29.17%

Facet	Level	Records	Detection	Pre-fault FAR
speed_by_torque	1800 / 80	24	66.67%	0.00%

S5.1 Fault-topology rows

Family	Terminals	Span (%)	Detection	Pre-fault FAR	First-window detection
turns	D01-D04	12.04	88.89%	33.33%	77.78%
turns	D02-D03	7.69	44.44%	44.44%	22.22%
turns	D05-D08	12.10	88.89%	33.33%	44.44%
turns	D06-D07	7.40	44.44%	33.33%	22.22%
turns	D09-D10	2.80	33.33%	22.22%	11.11%
turns	D11-D12	2.70	22.22%	44.44%	22.22%
turns	D13-D16	11.60	88.89%	11.11%	66.67%
turns	D14-D15	7.40	22.22%	22.22%	11.11%
turns	D17-D20	11.60	88.89%	0.00%	22.22%
turns	D18-D19	7.40	44.44%	33.33%	22.22%
turns	D21-D22	3.80	33.33%	44.44%	22.22%
turns	D23-D24	2.80	44.44%	44.44%	33.33%
windings	D01-D06	14.34	100.00%	33.33%	88.89%
windings	D01-D07	21.74	100.00%	11.11%	100.00%
windings	D04-D06	2.30	77.78%	77.78%	77.78%
windings	D04-D07	9.70	100.00%	44.44%	77.78%
windings	D11-D21	36.40	100.00%	33.33%	100.00%
windings	D11-D22	40.20	100.00%	44.44%	100.00%
windings	D12-D21	33.70	100.00%	11.11%	100.00%
windings	D12-D22	37.50	100.00%	0.00%	100.00%
windings	D13-D18	14.80	100.00%	33.33%	88.89%
windings	D13-D19	22.20	100.00%	22.22%	100.00%
windings	D16-D18	3.20	33.33%	55.56%	22.22%
windings	D16-D19	10.60	77.78%	55.56%	77.78%

These analyses were specified only after the independent PMSG result was known. They cannot replace confirmation.

Analysis	False alarms	FAR	Detected	Detection	First/second/censored
Session anchor	17/216	7.87%	167/216	77.31%	149/18/49
Matched-session topology cross-fit	14/216	6.48%	153/216	70.83%	133/20/63
Conditioned-anchor topology cross-fit	18/216	8.33%	154/216	71.30%	136/18/62

S6.1 Topology-disjoint fold results

Analysis	Fold	Threshold	False alarms	FAR	Detected	Detection
Matched anchor	1	35.1490	7	9.72%	62	86.11%
Matched anchor	2	88.0753	1	1.39%	39	54.17%
Matched anchor	3	53.9253	6	8.33%	52	72.22%
Conditioned anchor	1	31.7989	11	15.28%	65	90.28%
Conditioned anchor	2	93.1328	1	1.39%	38	52.78%
Conditioned anchor	3	49.5592	6	8.33%	51	70.83%

Run the feature/container audit, development benchmark, confirmation analysis, three session analyses, figure builder, and evidence validator in the order listed in README_REPRODUCE.md. The evidence validator checks seven immutable hashes, all headline numerators/denominators, the complete 3x3 condition grid, threshold instability, manuscript citations, and all ten figure files.

The following restrictions apply to every table and figure:

- two physical machines do not establish a fleet-population result;
- the 216 PMSG files are sessions/conditions, not independent machines;
- Wilson intervals are descriptive under dependence;
- motor-to-generator, topology, acquisition, and laboratory changes are confounded;
- post-reveal session repairs are internal mechanism analyses;
- censoring beyond 0.4 s is not proof of no later alarm;
- external abstention was zero, so context support did not detect conditional feature shift.

File	SHA-256
papers\paper3_calibration_transport\manuscript.md	6ef3f4991a0ec030b7b34faddfa0c4dec92bfcdb489aa32b9abc46cd8caa12c3
docs\paper3_reveal_log.md	daf4b650bfc003b09655a8f88715eccd3ae39e25679880debd2988f0c5dd501b
docs\paper3_literature_gap.md	0bdb6d3b341490c5ff9e8c799f1be7b1f60d7bf919b027d22a5af8203b89d52d
results\paper3_development\aggregate_summary.csv	39b861d62743575e057054cb54e4ef054e0bbe9cc6fab8ba31a2d6e2786f29b
results\paper3_pmsg_confirmation\aggregate_summary.csv	20e3cdbf90b9a3fa4927cb0ab5cdcdc018e52f834f223a519b023ef6ce2ba27a
results\paper3_pmsg_session_anchor\summary.json	c9ec7187f8a25c7bbdad42ce98c2cabe528a309f4f63aa591ee7100c07ffd53
results\paper3_pmsg_topology_crossfit\summary.json	d78a2ce955085e5ef373e87330e5d69213729d2823921c6117cffb409b7a78e8
results\paper3_pmsg_conditioned_anchor\summary.json	289f202a8ca4349cdf8f51a30a6aedf604caece480e49affd8e4402c8ca8992